CRIS
making IT happen

Project Internship Presentation
Centre for Railway Information Systems (CRIS)

Integrated Crew Deployment and Operational Capacity Management Framework

VAIBHAV PORWAL
(Roll No. : 225/UCS/066)
B. Tech. CSE
4th Year, Semester- VIII

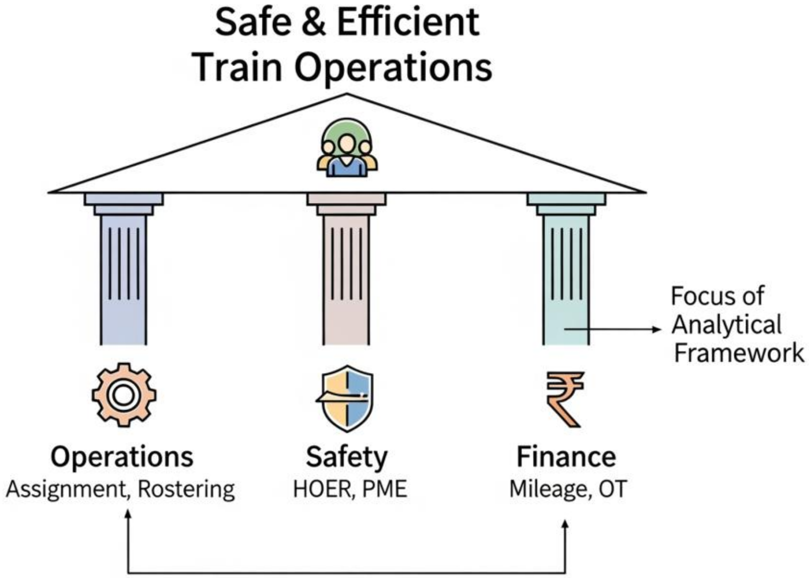
Project Mentored by
Avneesh Dwivedi (PPE/CMS), CRIS – Indian Railways

Organisation Profile - CRIS

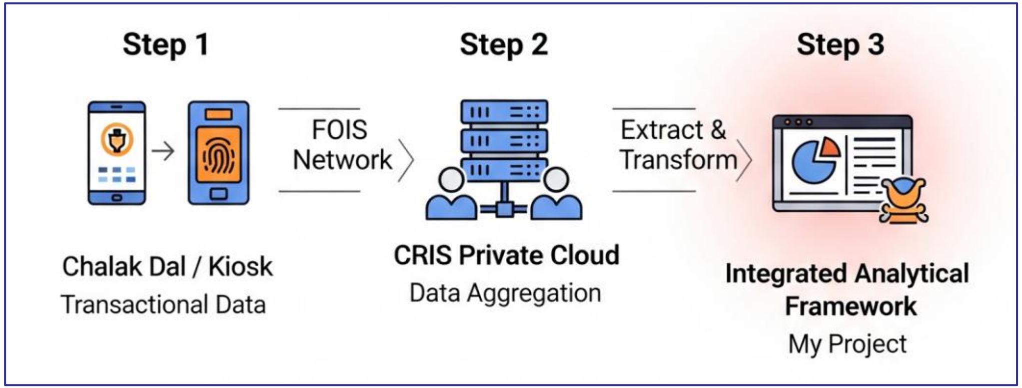
Organisation Type	An organisation under Ministry of Railways
Establishment	July 1986 (39 years ago)
Headquarters	Chanakya Rail Enclave, New Delhi, India
Core Competence	Design, development, and maintenance of Mission-Critical Railway IT Systems
Sector	IT Services, Systems Engineering, Data Infrastructure
Operational Scope	Pan-India
Workforce	~1000+ technical, engineering, and operations staff
Popular Products & Projects	RailOne, NTES, PRS, UTS, IRCTC RailConnect, RTIS, COA, FOIS, SLAM, CMS, PMS, ICMS, TMS, TDMS, RSMS, RailDrishti etc.

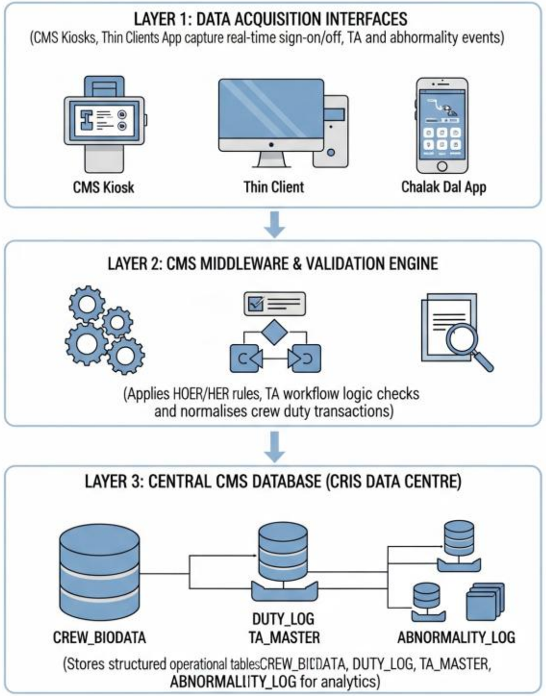
Crew Management System Group

- Manages **100,000+ Running Staff** (Drivers & Guards) 24x7 across **372+ Crew Lobbies** nationwide.
- **Criticality**: Ensures medically fit & rested pilots operate all trains.
- **Operational Efficiency**: Optimizes crew utilization and assignment; Eliminates manual "Call Boy" system via SMS/App.
- **Safety Compliance**: Monitors HOER (Hours of Employment Rules), Road Learning, and PME (Periodic Medical Exam) schedules.
- **Financial Control**: Precisely calculates Kilometer Allowance, monitors Overtime (OT), and prevents malingering.



System Architecture & Data Flow





Acquisition Layer (The Edge):

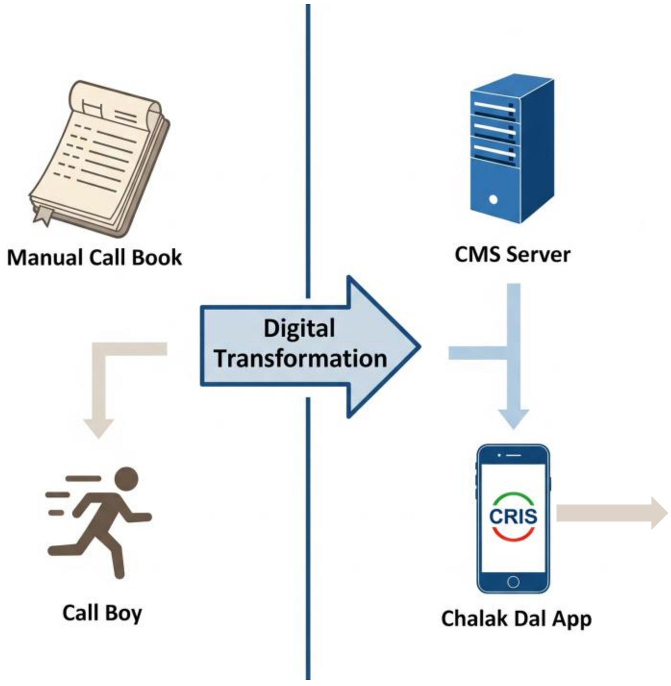
- **'Chalak Dal' App:** For Crew Sign On/Off and Abnormality Reporting.
- **Lobby Kiosks:** Biometric Authentication.

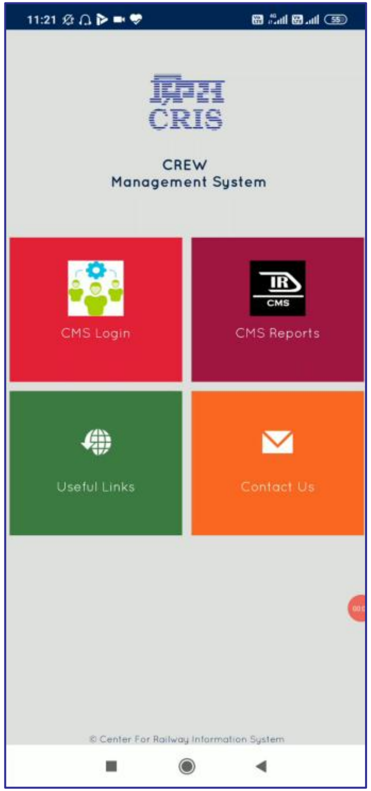
Transmission Layer: **FOIS Network:** Secure Wide Area Network (WAN).

Processing Layer (The Core): **CRIS Data Centre (CDC):** Central Database (OLTP) to Cloud Analytics.

Data Acquisition: Chalak Dal Interface

- **The Shift:** Transition from manual "Call Book" systems to Digital Notification.
- **'Chalak Dal' App:** The primary mobile interface for Running Staff (Pilots & Guards).
- **Crew Booking:** SMS-based call service eliminates physical "Call Boys".
- **Real-Time Sync:** Instant updates to the central server upon acknowledgement.

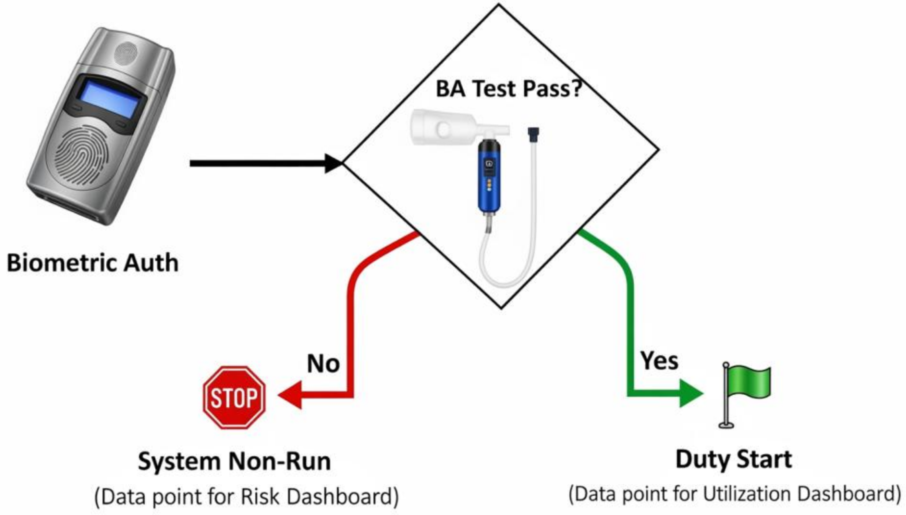


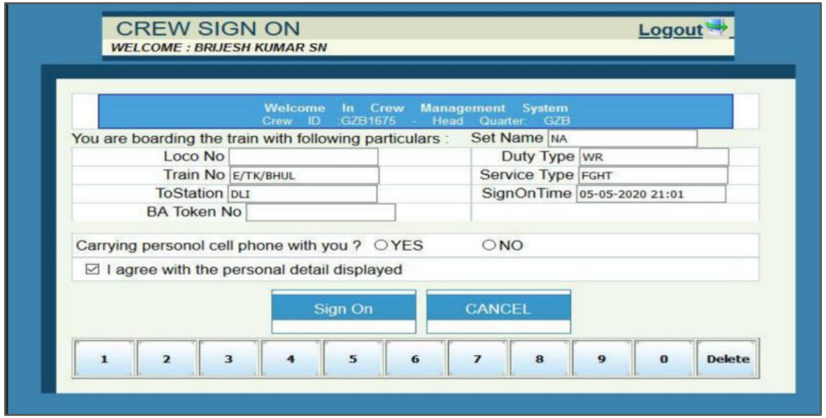


'Sign On' Transaction (Duty Start)

The Gatekeeper: The "Sign On" event is the single most critical data point. Mandatory Checks:

- **Biometric Auth**: Verifies identity against the Biodata master.
- **Breath Analyzer (BA) Test**: Mandatory safety check; failure means immediate "System Non-Run."
- **Read Circulars**: Ensures crew is updated on safety notices.



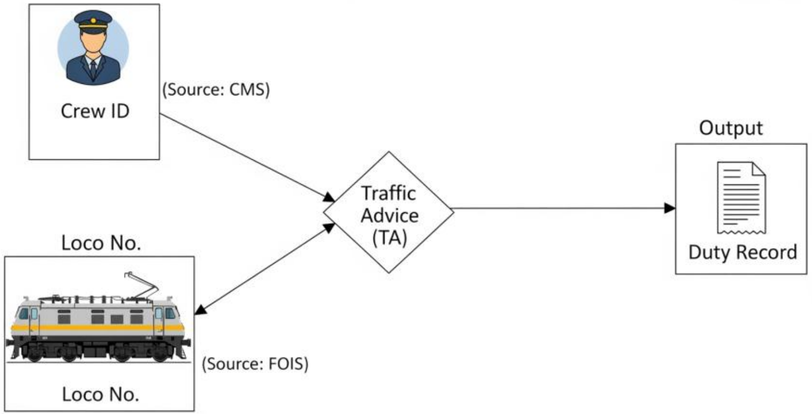


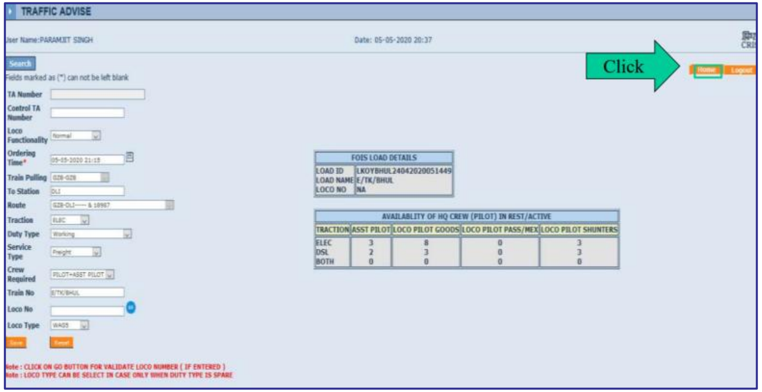
Traffic Advice (TA) : Operational Linking

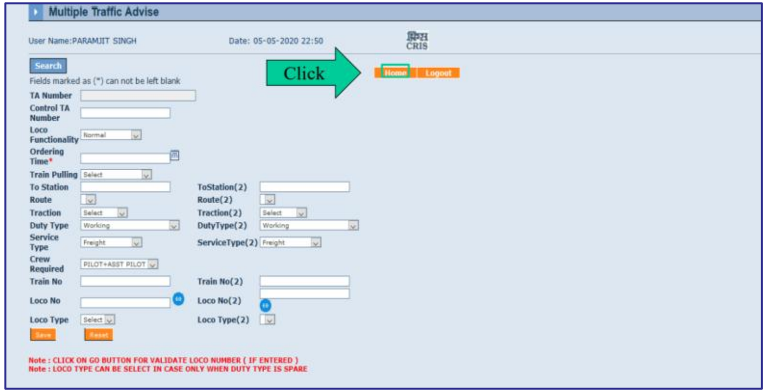
TA : The linkage between **Human Resources (Crew)** and **Physical Assets (Trains)**. The TA Process comprises:

- Supervisor selects a **Crew ID**.
- System fetches **Live Train/Loco Data** from **FOIS**
- Result: A mapped duty record (**Crew + Loco + Route**).

Analytical Value : This data allows calculation of "**Kilometers Earned**" per specific loco type.





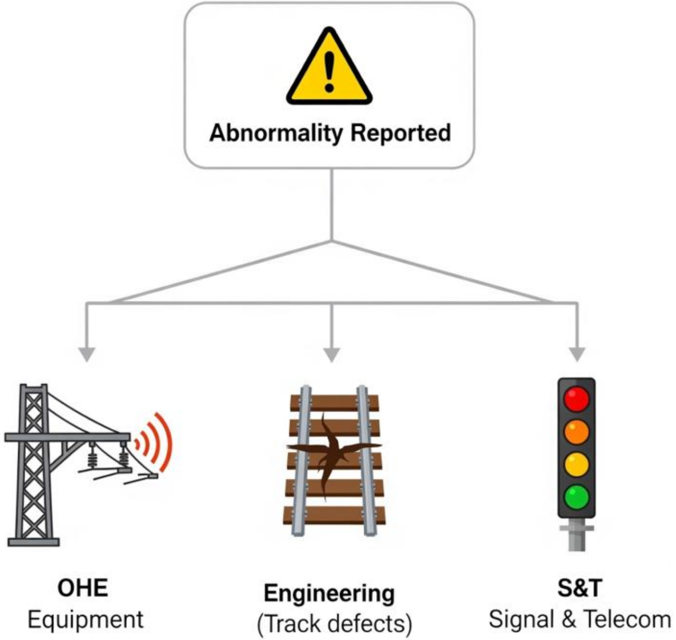


Abnormality Reporting

Crowd-sourced safety intelligence from 100,000+ crew members. Security Workflow includes:

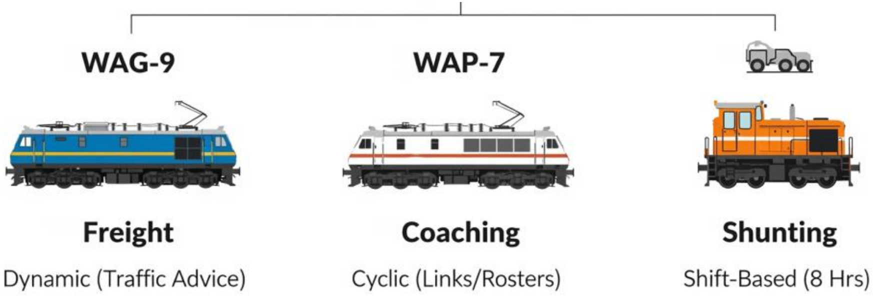
- Crew spots defect (e.g., Signal failure, OHE spark).
- Reports via App/Kiosk under specific **Sub-heads** (OHE, S&T, Engineering).
- Alert sent to concerned Department & Control.

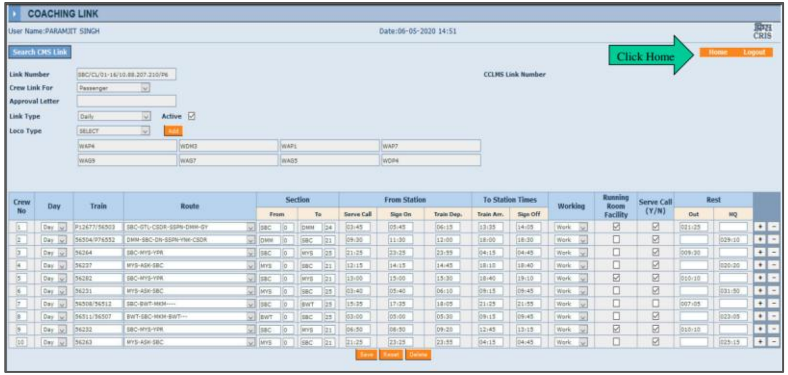
Dashboard Input: Visualizing "Open vs. Closed" abnormalities.



Coaching & Shunting Operations

- **Coaching Operations**: Unlike Freight, Coaching uses a fixed "Link" (Roster) system (Daily/Weekly cycles). Crew are assigned to specific trains (e.g., Vande Bharat/Shatabdi) in advance.
- **Shunting Operations**:
  - Yard duties are 8-hour fixed shifts.
  - Managed via "Shunter Rosters," not trip-based TAs.
- **Analytical Challenge**: Normalizing these fixed-roster hours against dynamic freight hours is required for a unified "Total Duty" metric.




COACHING LINK

User Name:PARAMJIT SINGH
Date:06-05-2020 14:51

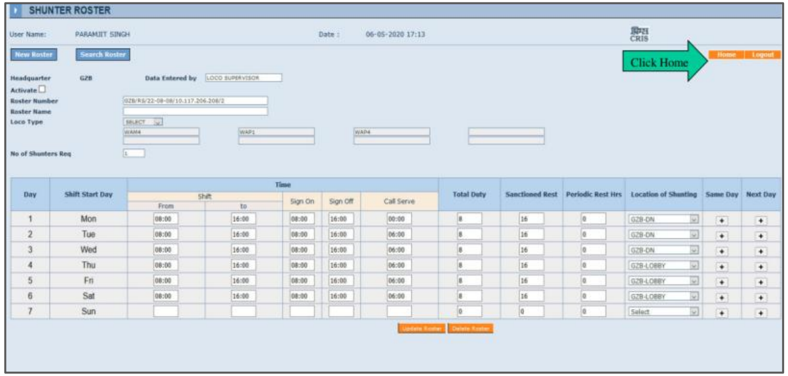
Search CMS Link
Click Home
Home Logout

Link Number
CCLMS Link Number
Crew Link For: Passenger
Approval Letter
Link Type: Daily   Active
Loco Type: SELECT   Add

WAP4 WDM3 WAP1 WAP7
WAG9 WAG7 WAG5 WDP4

Crew No	Day	Train	Route	Section From		Section To		From Station Serve Call	From Station Sign On	From Station Train Dep.	To Station Times Train Arr.	To Station Times Sign Off	Working	Running Room Facility	Serve Call (Y/N)	Rest Out	Rest HQ
1	Day	P12677/56503	SBC-GTL-CSDR-SSPN-DMM-GY	SBC	0	DMM	24	03:45	05:45	06:15	13:35	14:05	Work	☑	☑	021:25	
2	Day	56504/P76552	DMM-SBC-DN-SSPN-YNK-CSDR	DMM	0	SBC	21	09:30	11:30	12:00	18:00	18:30	Work	☐	☑		029:10
3	Day	56264	SBC-MYS-YPR	SBC	0	MYS	25	21:25	23:25	23:55	04:15	04:45	Work	☐	☑	009:30	
4	Day	56237	MYS-ASK-SBC	MYS	0	SBC	21	12:15	14:15	14:45	18:10	18:40	Work	☐	☑		020:20
5	Day	56282	SBC-MYS-YPR	SBC	0	MYS	21	13:00	15:00	15:30	18:40	19:10	Work	☑	☑	010:10	
6	Day	56231	MYS-ASK-SBC	MYS	0	SBC	25	03:40	05:40	06:10	09:15	09:45	Work	☐	☑		031:50
7	Day	56508/56512	SBC-BWT-MKM----	SBC	0	BWT	25	15:35	17:35	18:05	21:25	21:55	Work	☐	☐	007:05	
8	Day	56511/56507	BWT-SBC-MKM-BWT---	BWT	0	SBC	25	03:00	05:00	05:30	09:15	09:45	Work	☐	☑		023:05
9	Day	56232	SBC-MYS-YPR	SBC	0	MYS	21	06:50	08:50	09:20	12:45	13:15	Work	☑	☑	010:10	
10	Day	56263	MYS-ASK-SBC	MYS	0	SBC	21	21:25	23:25	23:55	04:15	04:45	Work	☐	☑		025:15

Save Reset Delete


SHUNTER ROSTER

User Name: PARAMJIT SINGH
Date : 06-05-2020 17:13

New Roster
Search Roster
Click Home
Home Logout

Headquarter: GZB
Data Entered by: LOCO SUPERVISOR
Active
Roster Number
Roster Name
Loco Type: SELECT
WAM4 WAP1 WAP4
No of Shunters Req: 1

Day	Shift Start Day	Time: Shift From	Time: Shift to	Time: Sign On	Time: Sign Off	Time: Call Serve	Total Duty	Sanctioned Rest	Periodic Rest Hrs	Location of Shunting	Same Day	Next Day
1	Mon	08:00	16:00	08:00	16:00	00:00	8	16	0	GZB-DN	+	+
2	Tue	08:00	16:00	08:00	16:00	06:00	8	16	0	GZB-DN	+	+
3	Wed	08:00	16:00	08:00	16:00	06:00	8	16	0	GZB-DN	+	+
4	Thu	08:00	16:00	08:00	16:00	06:00	8	16	0	GZB-LOBBY	+	+
5	Fri	08:00	16:00	08:00	16:00	06:00	8	16	0	GZB-LOBBY	+	+
6	Sat	08:00	16:00	08:00	16:00	06:00	8	16	0	GZB-LOBBY	+	+
7	Sun						0	0	0	Select	+	+

Update Roster Delete Roster

'System Non-Run' Concept

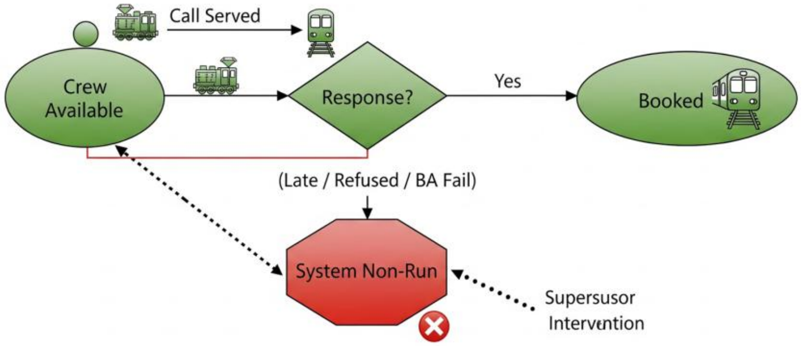
A system-generated status when a crew member fails to perform an assigned duty.

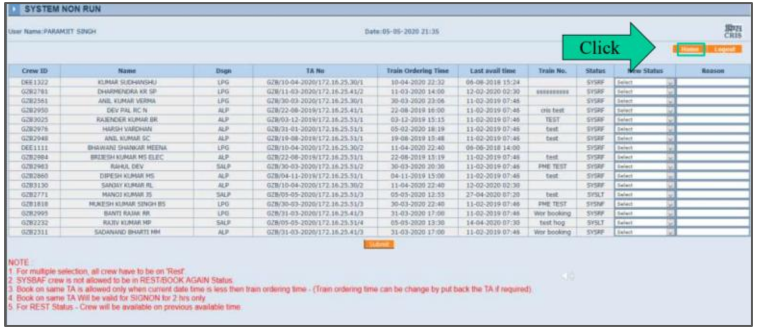
Trigger Events:

- **Late Arrival**: Crew fails to sign on within the stipulated time.
- **Refusal**: Crew refuses the call.
- **BA Failure**: Crew fails the Breath Analyzer test.

Resolution: Requires Supervisor intervention to regularize status.

Metric Utility: A key indicator of Crew Discipline and Availability Wastage.





SYSTEM NON RUN

User Name:PARAMJIT SINGH

Date:05-05-2020 21:35

Click

Crew ID	Name	Dsgn	TA No	Train Ordering Time	Last avail time	Train No.	Status	New Status	Reason
DEE1322	KUMAR SUDHANSHU	LPG	GZB/10-04-2020/172.16.25.30/1	10-04-2020 22:32	06-08-2018 15:24		SYSRF	Select	
GZB2781	DHARMENDRA KR SP	LPG	GZB/11-03-2020/172.16.25.41/2	11-03-2020 14:00	12-02-2020 02:30	##########	SYSRF	Select	
GZB2561	ANIL KUMAR VERMA	LPG	GZB/30-03-2020/172.16.25.30/1	30-03-2020 23:06	11-02-2019 07:46		SYSRF	Select	
GZB2950	DEV PAL RC N	ALP	GZB/22-08-2019/172.16.25.41/1	22-08-2019 16:00	11-02-2019 07:46	cris test	SYSRF	Select	
GZB3025	RAJENDER KUMAR BR	ALP	GZB/03-12-2019/172.16.25.51/1	03-12-2019 15:15	11-02-2019 07:46	TEST	SYSRF	Select	
GZB2976	HARSH VARDHAN	ALP	GZB/31-01-2020/172.16.25.51/1	05-02-2020 18:19	11-02-2019 07:46	test	SYSRF	Select	
GZB2948	ANIL KUMAR SC	ALP	GZB/19-08-2019/172.16.25.51/1	19-08-2019 15:48	11-02-2019 07:46	test	SYSRF	Select	
DEE1111	BHAWANI SHANKAR MEENA	LPG	GZB/10-04-2020/172.16.25.30/2	11-04-2020 22:40	06-08-2018 14:00		SYSRF	Select	
GZB2984	BRIJESH KUMAR MS ELEC	ALP	GZB/22-08-2019/172.16.25.51/1	22-08-2019 13:19	11-02-2019 07:46	test	SYSRF	Select	
GZB2983	RAHUL DEV	SALP	GZB/30-03-2020/172.16.25.51/2	30-03-2020 20:30	11-02-2019 07:46	PME TEST	SYSRF	Select	
GZB2860	DIPESH KUMAR MS	ALP	GZB/04-11-2019/172.16.25.51/1	04-11-2019 15:00	11-02-2019 07:46	test	SYSRF	Select	
GZB3130	SANJAY KUMAR RL	ALP	GZB/10-04-2020/172.16.25.30/2	11-04-2020 22:40	12-02-2020 02:30		SYSRF	Select	
GZB2771	MANOJ KUMAR JS	SALP	GZB/05-05-2020/172.16.25.51/3	05-05-2020 12:55	27-04-2020 07:20	test	SYSLT	Select	
GZB1818	MUKESH KUMAR SINGH BS	LPG	GZB/30-03-2020/172.16.25.51/3	30-03-2020 22:40	11-02-2019 07:46	PME TEST	SYSNF	Select	
GZB2995	BANTI RAJAK RR	LPG	GZB/31-03-2020/172.16.25.41/3	31-03-2020 17:00	11-02-2019 07:46	Wer booking	SYSRF	Select	
GZB2232	RAJIV KUMAR MP	SALP	GZB/05-05-2020/172.16.25.51/4	05-05-2020 13:30	14-04-2020 07:30	test hog	SYSLT	Select	
GZB2311	SADANAND BHARTI MM	ALP	GZB/31-03-2020/172.16.25.41/3	31-03-2020 17:00	11-02-2019 07:46	Wer booking	SYSRF	Select	

NOTE

1. For multiple selection, all crew have to be on 'Rest'
2. SYSBAF crew is not allowed to be in REST/BOOK AGAIN Status
3. Book on same TA is allowed only when current date time is less then train ordering time - (Train ordering time can be change by put back the TA if required)
4. Book on same TA Will be valid for SIGNON for 2 hrs only
5. For REST Status - Crew will be available on previous available time

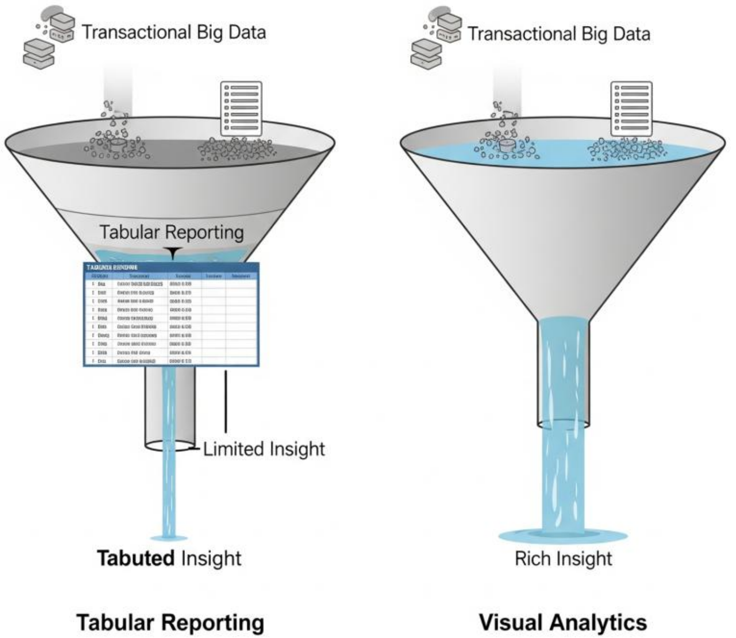
Problem Statement : Analytical Gap

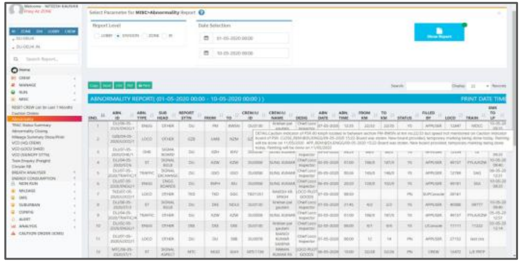
Data Rich, Insight Poor: CMS captures 100% of transactions but presents static data.

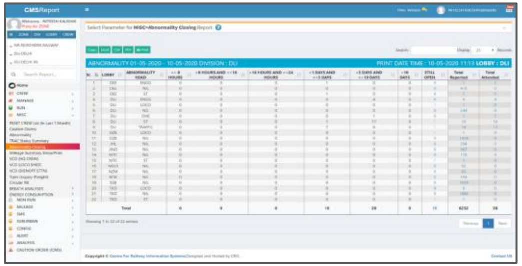
Fragmented View:

- Manual correlation of Biodata (Static) and Mileage (Dynamic) is required.
- No "Single Pane of Glass" for Zonal trends (e.g., Fatigue vs. Traction).

Consequence: Reactive decisions instead of proactive resource planning.







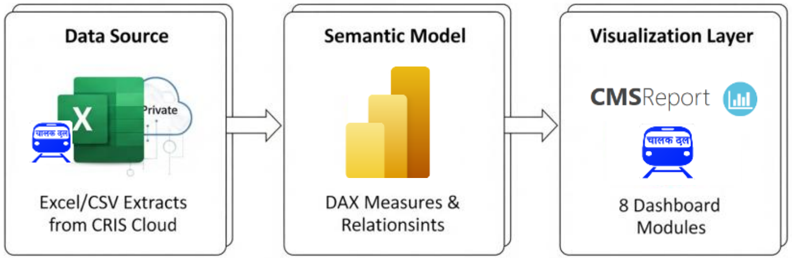
Solution : Integrated Analytical Framework

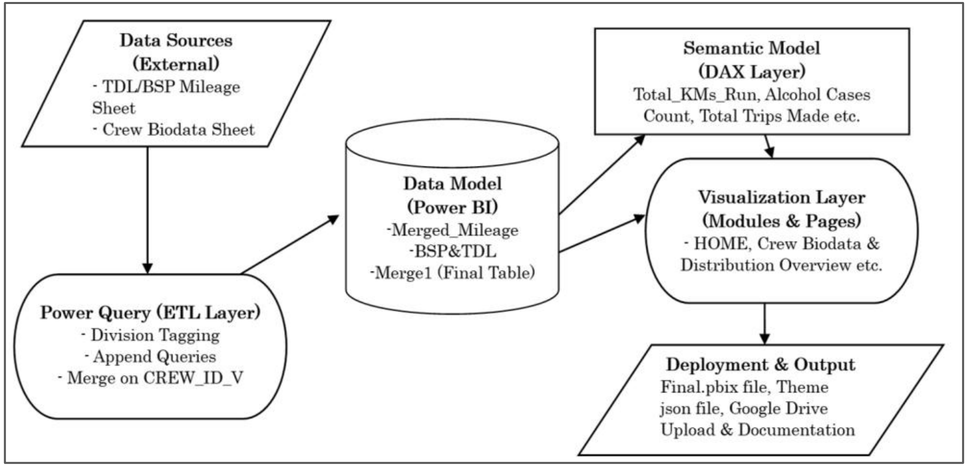
The Solution: An Integrated Analytical Layer built on Power BI.

Core Strategy:

- Ingest: Mileage Data + Crew Biodata.
- Model: Star Schema integrating diverse datasets (Freight, Coaching, Safety).
- Visualize: 8 Modular Dashboards for "Macro-Level Monitoring".

Deployment Target: Deployed dashboards on 'Chalak Dal Reports' Module.

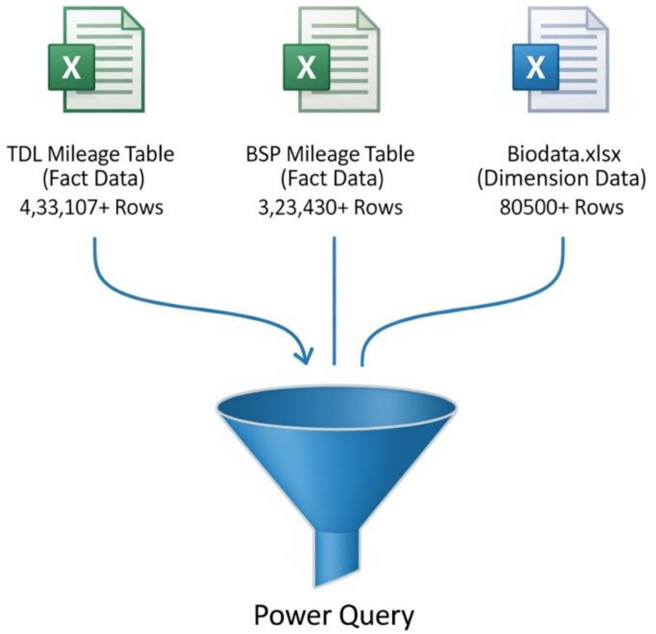




Aggregated Data: Source for Dashboards

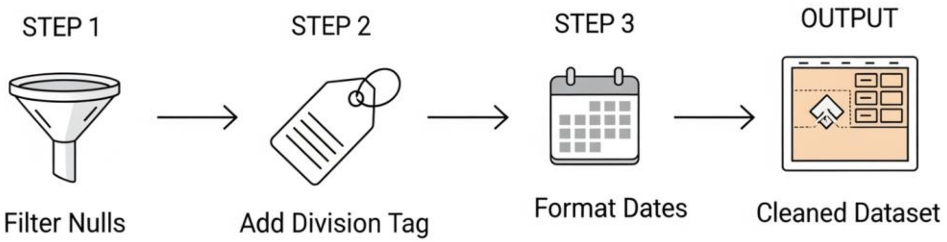
1. **Crew Mileage Data :** Dynamic, Transactional
2. **Crew Biodata :** Static, Dimension Table

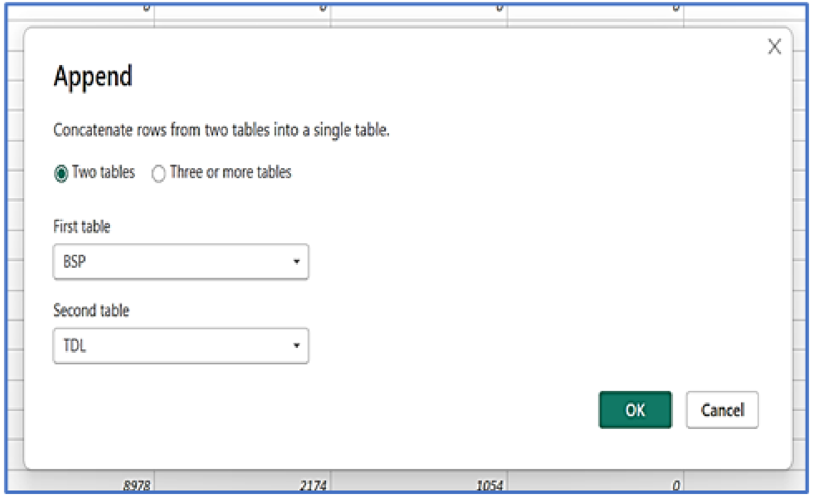
File Name	Sheet	Table Description	Key Columns
1_TDL_BSP_5Month_MILEAGE_DATA.xlsx	TDL (433107+ Rows)	Contains 5-month crew mileage and operational data for TDL.	CREW_ID_V, TOTAL_DUTY, NON_OFF_KMS, OFF1_KMS, OFF2_KMS, ALKM_NON_LEAVE, ALKM_LEAVE, NRDA_KMS, OSRA_KMS, TOTAL_KMS, RRA, BOR, NGHT, NH, SHUNT_COUNT, TRIP_COUNT, SLOT_NUMBER_N, TENTATIVE_FLAG, DATE_TIME_D, RUN_DUTY_MIN, NON_RUN_DUTY_MIN, FOOT_PLT_KM, ABSENT, SICK_LEAVE, LEAVE_DAYS, STATIONAY_DUTY, TEST_TRNG, OTHER_NON_LEAVE, CREW_BASE_ID_V, SPARE_KMS_N, SPARE_DUTY_MINS_N, NO_OF_TRIPS_N, NH_DATES, HQ_CODE_C, EMP_NO_V, COACH_RUN_DUTY_MIN_N, COACH_FOOT_PLT_KM_N
	BSP (3,23,432+ Rows)	Contains 5-month crew mileage and operational data for BSP.	
1_TDL_BSP_Crew_Biodata.xlsx	BSP&TDL (80,500 + Rows)	Contains crew biodata across both BSP and TDL bases/lobbies	NAME_V, HQ_CODE_C, ORG_TYPE_C, CREW_ID_V, CREW_DESIG_V, INACTIVE_STTS_V, CREW_CADRE_V, TRCTN_C, EMP_NO_V, PF_CODE_N, INACTIVE_RESN_V, MOBILE_NO_N, CREW_BASE_ID_V, IPAS_FLAG_C, AU_CODE_V, ALCOHOL_C, FLAG_C, VALID_FROM_DATETIME_D, VALID_TO_DATETIME_D, LI_ID_V



Data Appending and Lobby Tagging

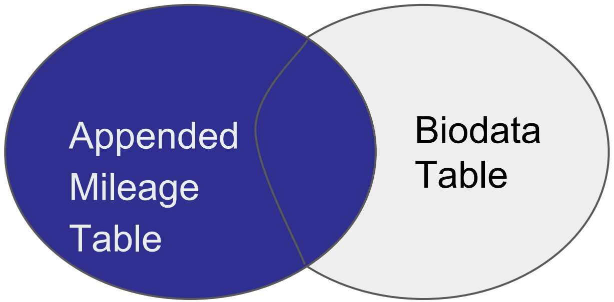
- **Cleaning**: Removing Null/Blank 'Sign Off' entries (System Non-Run).
- **Tagging**: Adding a custom [Division] column to distinguish TDL vs. BSP data.
- **Standardization**: Formatting Date/Time columns for Time Intelligence functions.



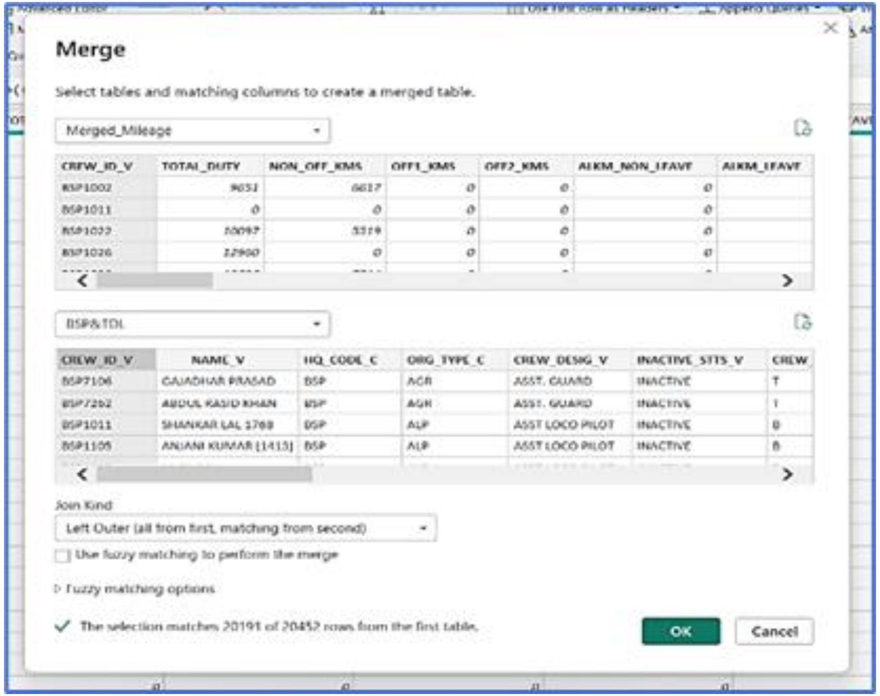


Creating the Master Table

- **The Challenge**: Correlating "Who" (Biodata) with "What" (Mileage).
- **The Solution**: Merging queries into a single Master Table.
- **Join Key:** CREW_ID (Unique Primary Key).
- **Join Type**: Left Outer Join (Keep all Mileage transactions, match with Biodata).



Joined on CREW_ID

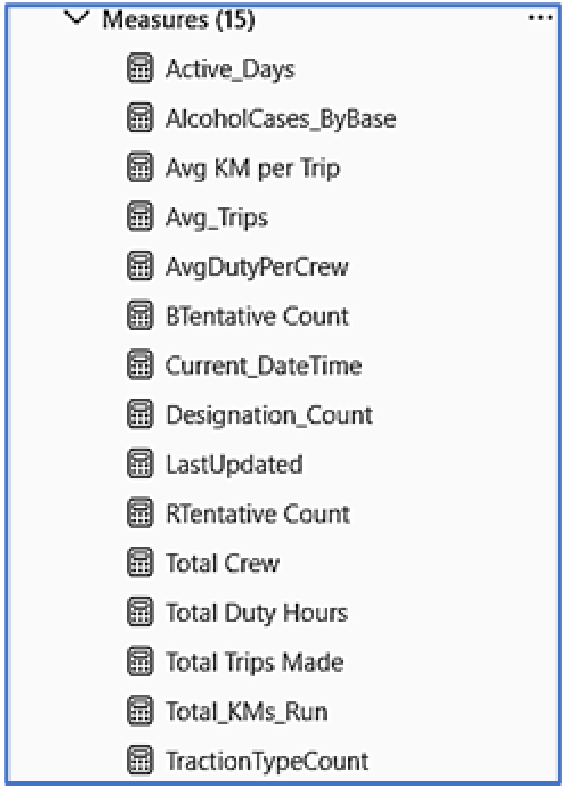


Measure Engineering & DAX Measures

The Logic Engine: Using DAX (Data Analysis Expressions) to create dynamic KPIs.

Why DAX? Allows measures to recalculate instantly when a user filters by Date or Lobby.

Measure Name	Purpose
Total Crew	Count of unique CREW_ID_V
Total Trips Made	Sum of all values in TRIP_COUNT
Total Duty Hours	Aggregate of TOTAL_DUTY
Total_KMs_Run	Sum of TOTAL_KMS
AlcoholCases_ByBase	Crew flagged with ALCOHOL_C = "Y"
Inactive Crew Count	Count of records with non-blank INACTIVE_STTS_V
AvgDutyPerCrew	Average duties across crew members
Avg KM per Trip	Total_KMS ÷ TRIP_COUNT where valid
BTentative Count	Count where TENTATIVE_FLAG = "B"
Current_DateTime	Current system date-time (for footer display)
Active Days	Count of distinct DATE_TIME_D

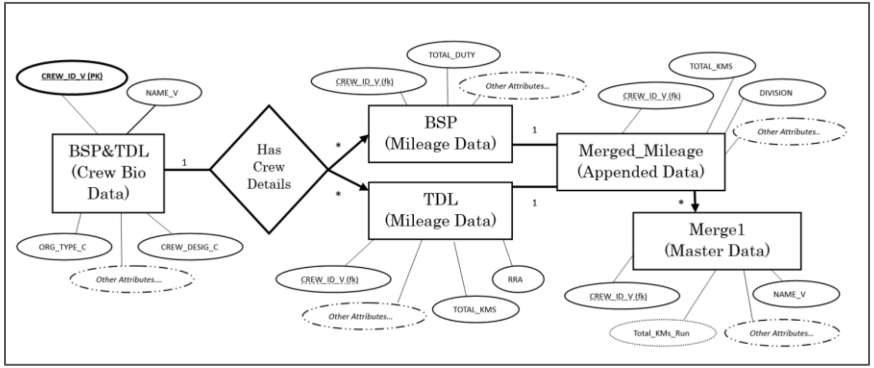


Final Data Model

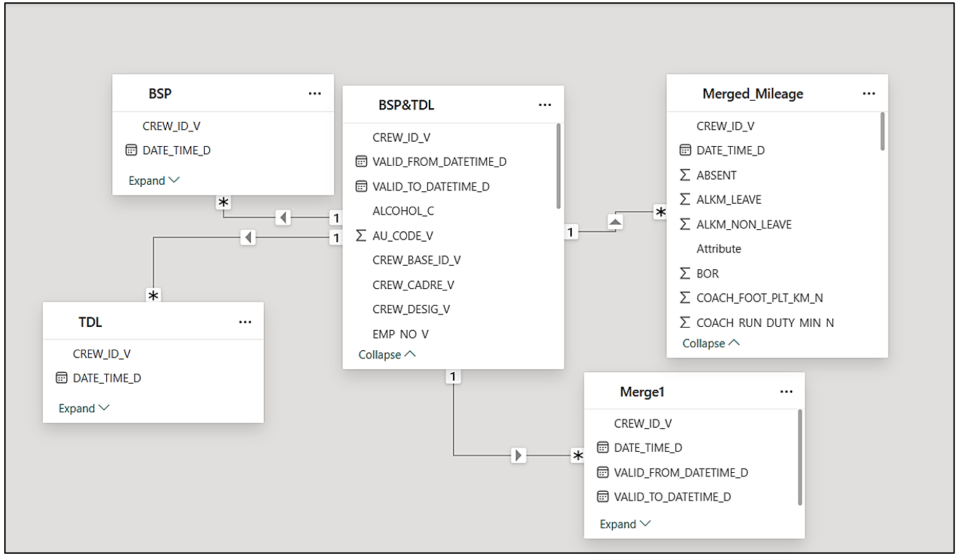
Final Merged Table contains **66M+ KMs** of running staff data and a robust data model scalable to every crew lobby and division.

Relationships (4)

- BSP[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
- Merge1[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
- Merged_Mileage[CREW_ID_V] <— BSP&TDL[CREW_ID_V]
- TDL[CREW_ID_V] <— BSP&TDL[CREW_ID_V]

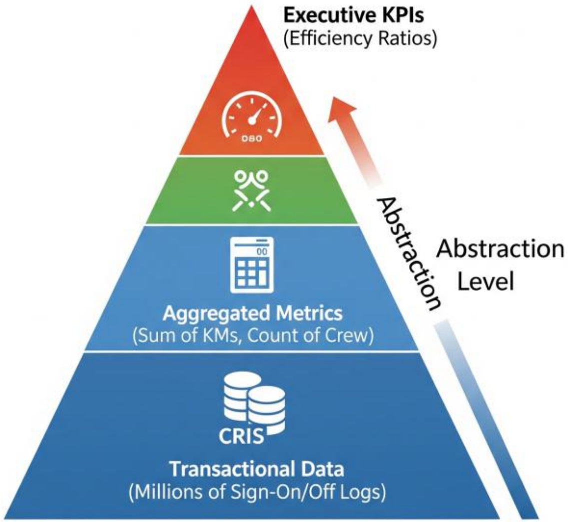


B.4 PowerBI Model and Schema View

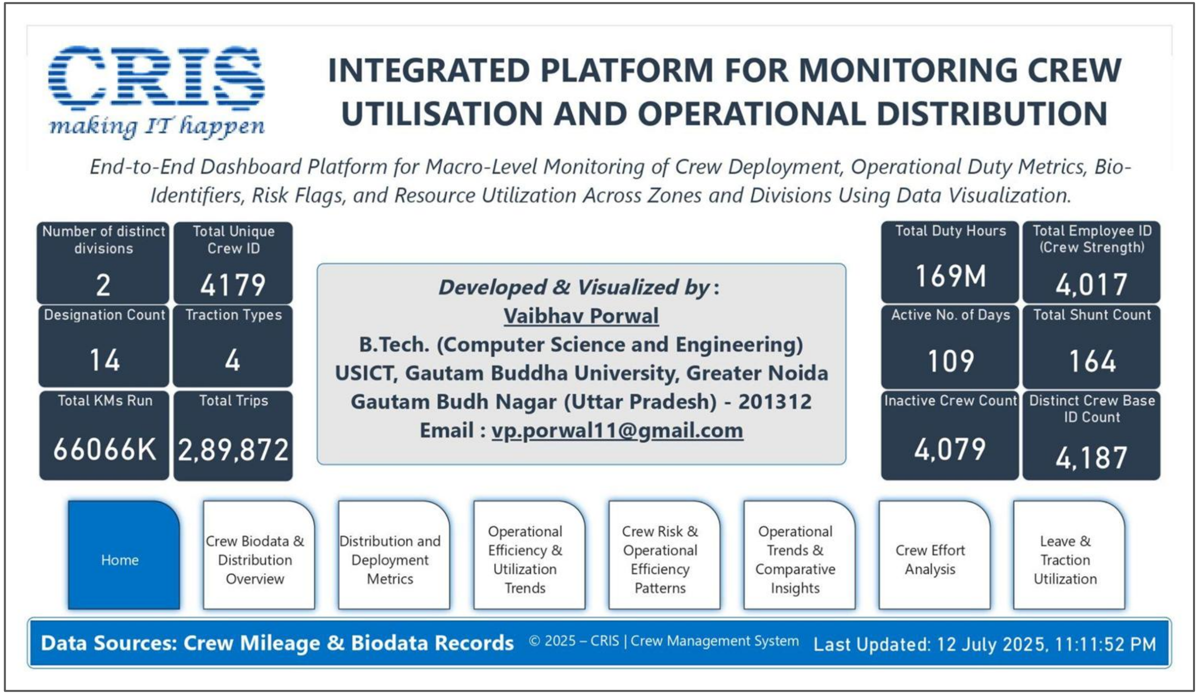


Crew Biodata & Competency Analysis

The "Cockpit" View: A single-screen summary for Zonal Managers (e.g., GM/Operations) with KPIs and navigation tiles.



Raw Data to Visual Intelligence



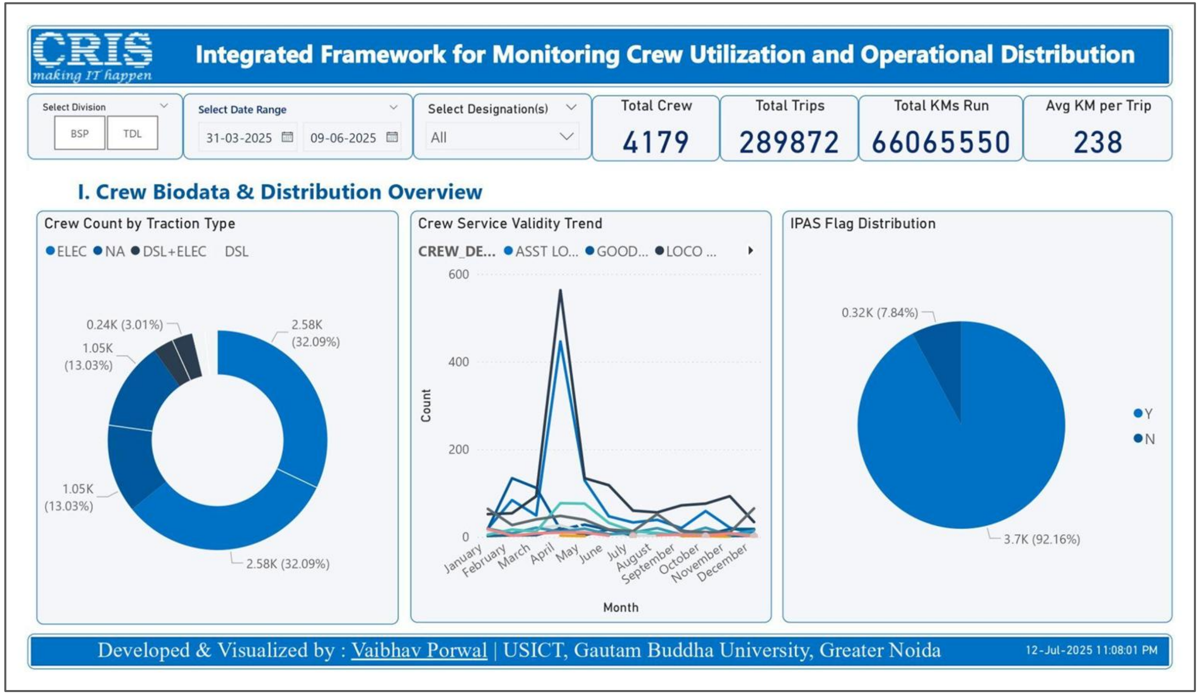
Crew Biodata & Competency Analysis

Analyzing crew distribution by Designation (ALP, LPG, LPM) and Traction (Electric/Diesel).

Competency Mapping:

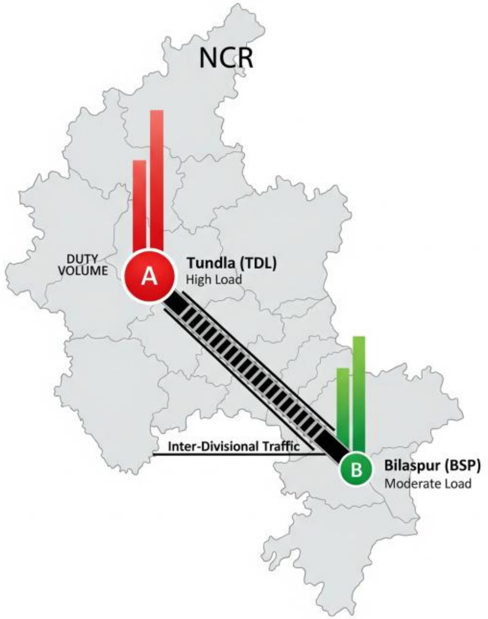
- Electric (E): Qualified for OHE routes (e.g., WAP-7).
- Diesel (D): Qualified for non-electrified routes (e.g., WDP-4).

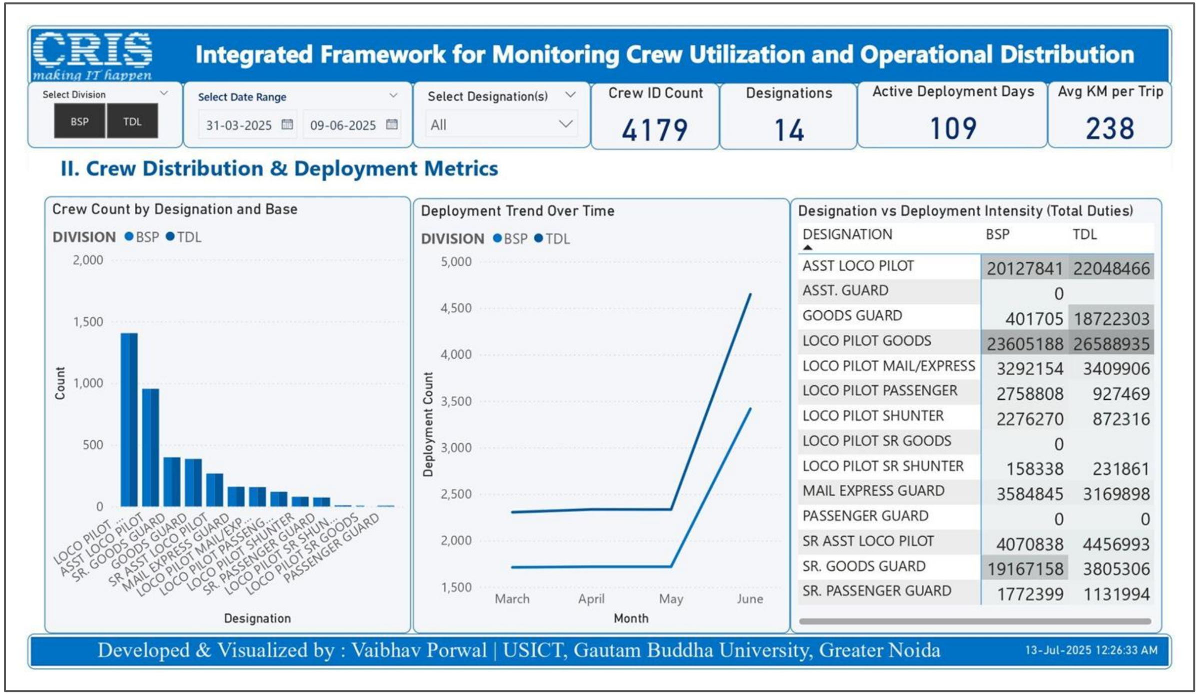
Strategic Utility: Identifying "Traction Mismatch" (e.g., Surplus Diesel crew in an Electrified zone).



Deployment & Operational Intensity

- **Spatial Analysis**: Compares workload between Crew Lobbies (Tundla vs. Bilaspur).
- **Designation Heatmap**: Identifies which rank (e.g., Loco Pilot Goods) carries max operational load.
- **Load Balancing:**





DESIGNATION	BSP	TDL
ASST LOCO PILOT	20127841	22048466
ASST. GUARD	0	
GOODS GUARD	401705	18722303
LOCO PILOT GOODS	23605188	26588935
LOCO PILOT MAIL/EXPRESS	3292154	3409906
LOCO PILOT PASSENGER	2758808	927469
LOCO PILOT SHUNTER	2276270	872316
LOCO PILOT SR GOODS	0	
LOCO PILOT SR SHUNTER	158338	231861
MAIL EXPRESS GUARD	3584845	3169898
PASSENGER GUARD	0	0
SR ASST LOCO PILOT	4070838	4456993
SR. GOODS GUARD	19167158	3805306
SR. PASSENGER GUARD	1772399	1131994

Operational Efficiency & Financial Impact

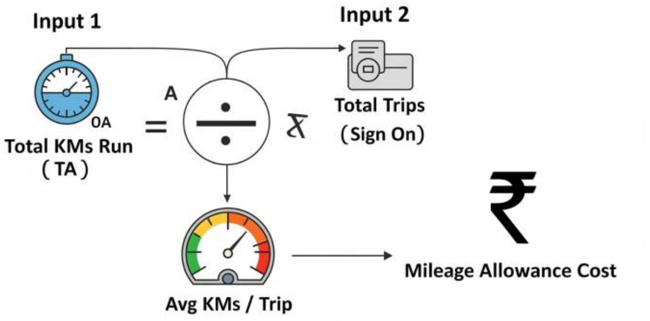
Kilometer Allowance (KMA):

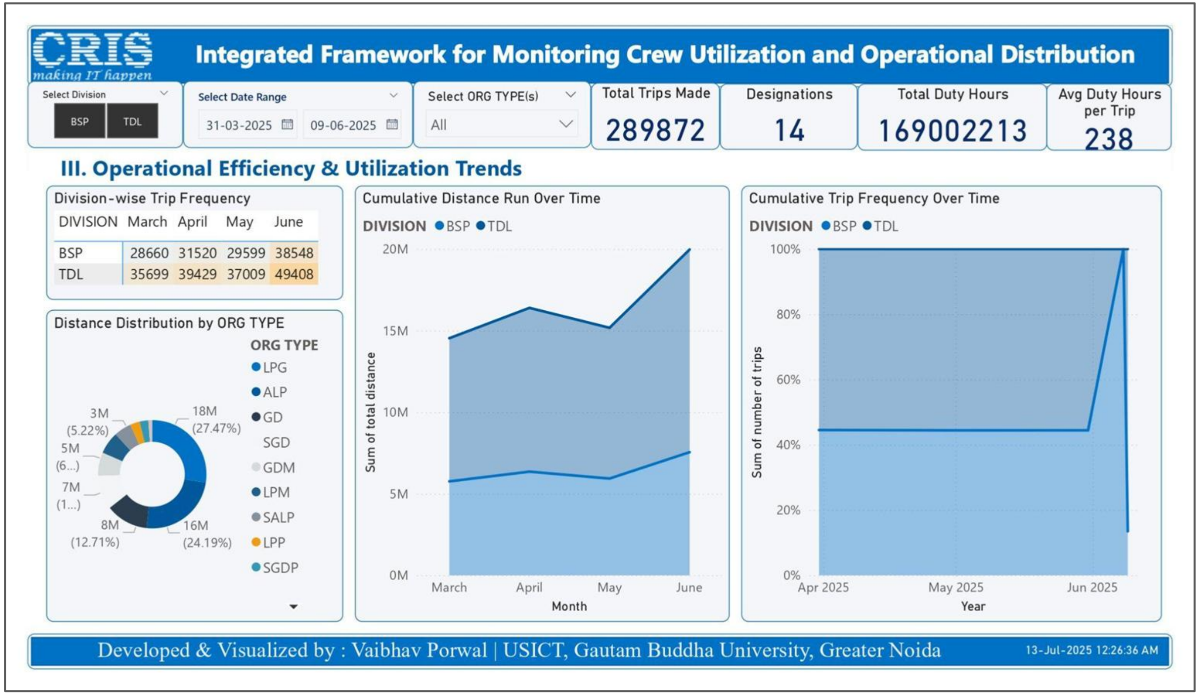
- Crew pay is based on KMs earned.
- Higher KMs indicate higher operational efficiency and cost.

KPIs:

- Avg KMs per Trip: Identifies long-haul vs. short shunting trips.
- Cumulative Distance: Total network coverage.

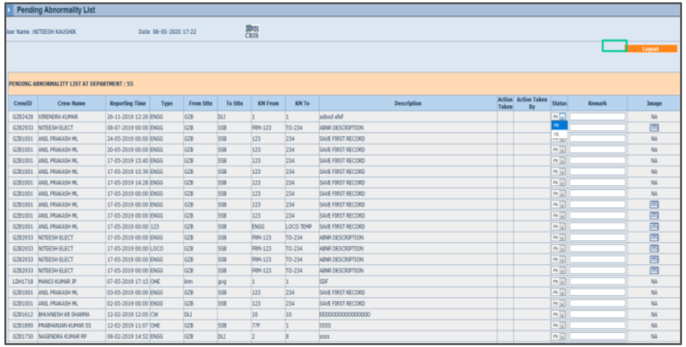
Trend Analysis: Correlate KMs run with seasonal traffic spikes.

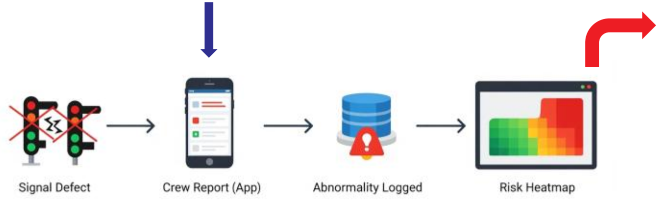


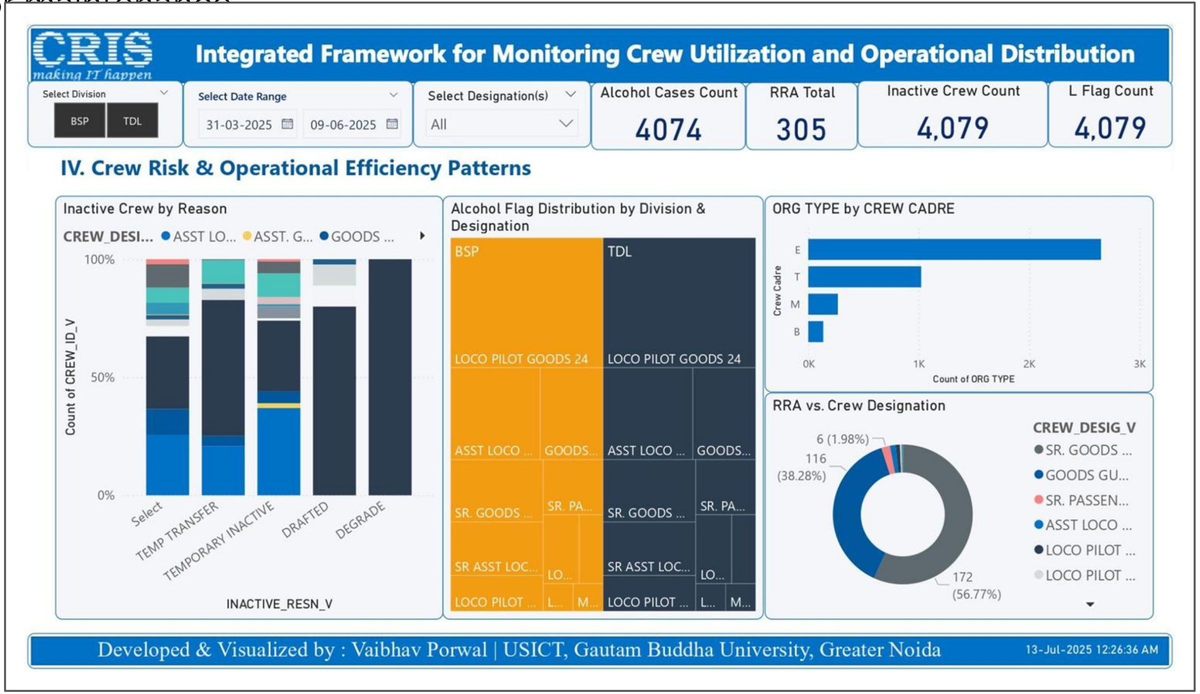


Risk & Efficiency (Abnormality Analysis)

- **Edge Source:** Abnormality Module (Chalak Dal App).
- **Signal Defects:** Reported by pilots via App.
- **Action**: Prioritizing high-risk zones for maintenance.

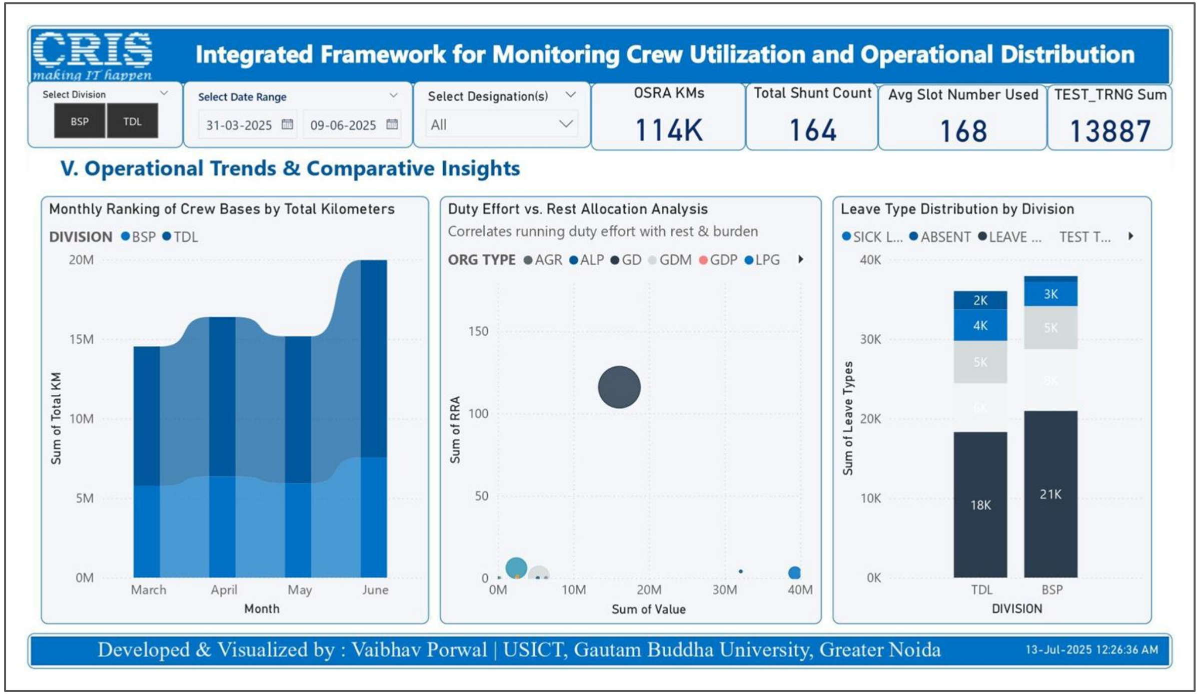






Rest & Fatigue Management

- **Regulatory Compliance**: Monitoring adherence to HOER 2005.
- **Critical Violation**: "Breach of Rest" (Sending crew out with < 12/16 hours rest).
- **Fatigue Analysis:**
    - Correlation: Heavy Duty (14+ Hrs) → Low Rest.
    - Safety Risk: Fatigued pilots are a primary cause of SPAD (Signal Passed at Danger).
- **Visual**: Plots identifying outliers (High Work / Low Rest).

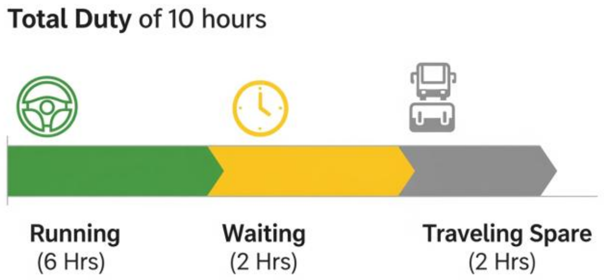


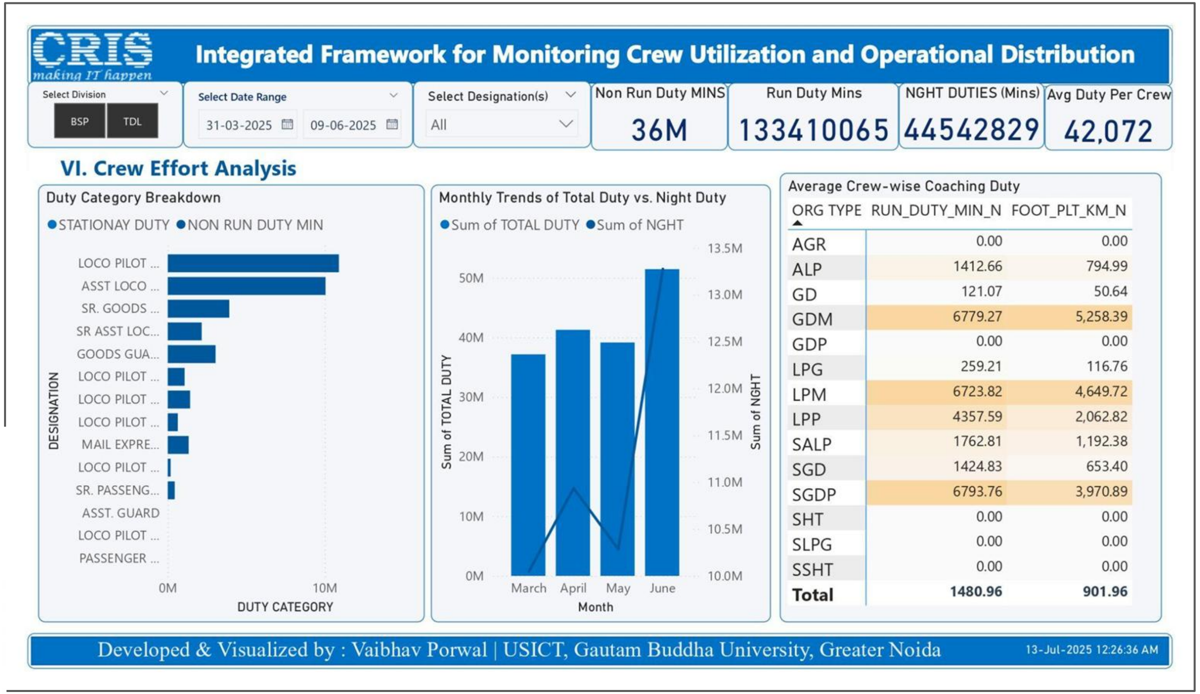
Crew Effort Analysis

Differentiating "**Productive**" vs. "**Unproductive**" time.

- **Running Duty**: Actual train driving.
- **Non-Running Duty**: Waiting, or traveling as a passenger (Spare).

Optimization Goal: Minimize "Non-Run" hours to reduce crew wastage.

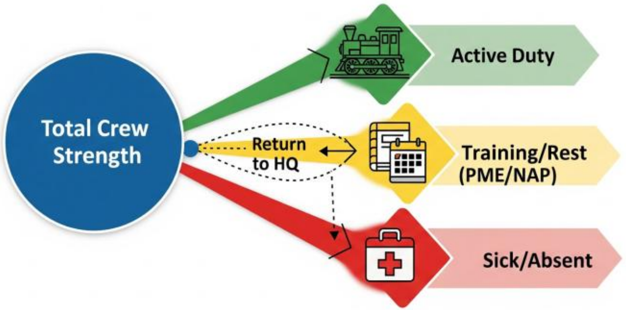


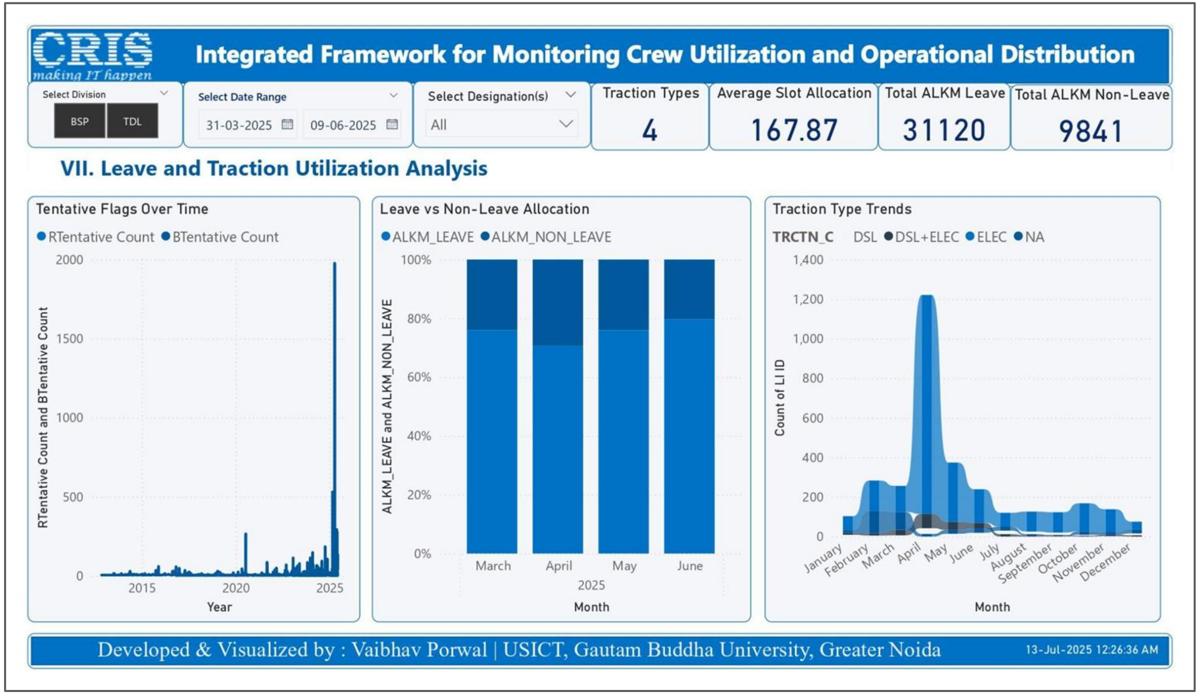


Leave and Traction Utilization Analysis

Tracking crew not available for duty to predict shortages in heavy traffic periods

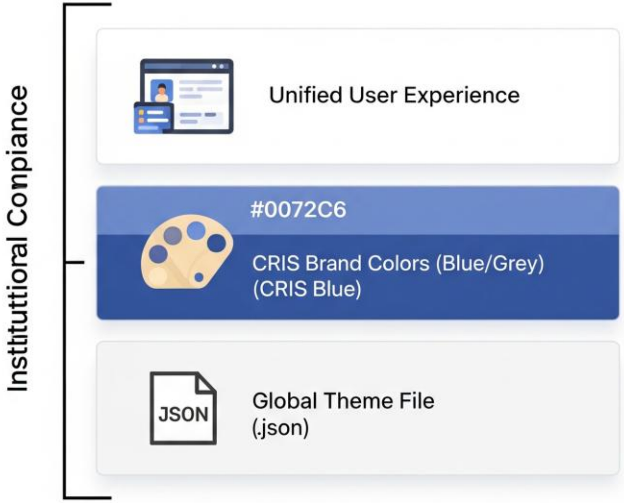
- **Leave Analysis:** Sick Leave vs. Casual Leave (NAP/LAP).
- **Tentative Flags**: Crew marked as "Tentative" (Uncertain availability).
- **Traction Trends**: Usage of Electric (E) vs. Diesel (D) traction over time





User Interface & CRIS Branding

- **Color Palette**: CRIS Blue (#0072C6) and Grey to match the 'Chalak Dal' App identity.
- **Typography**: Segoe UI (Standard Windows/Office font used by CRIS).
- **Developed** a reusable CRIS_Theme.json file to ensure all future reports maintain branding.



{"name":"CRIS Theme","textClasses":{"label":{"color":"#2C3E50","fontSize":12}},"dataColors":["#0072C6","#005A9E","#2C3E50","#F4F6F7","#D5DBDB","#1F618D","#85929E","#F39C12","#
3599B8","#DFBFBF","#4AC5BB","#5F6B6D","#FB8281","#F4D25A","#7F898A","#A4DDEE","#FDAB89","#B687AC","#28738A","#A78F8F","#168980","#293537","#BB4A4A","#B59525","#475052","#6A9FB0","#BD7150","#
7B4F71","#1B4D5C","#706060","#0F5C55","#1C2325"],"bad":"#C0392B","neutral":"#5D6D7E","good":"#27AE60","minimum":"#D6EAF8","center":"#85C1E9","maximum":"#
154360","visualStyles":{"*":{"*":{"outspacePane":[{"backgroundColor":{"solid":{"color":"#EAF2F8"}},"foregroundColor":{"solid":{"color":"#
1A5276"}},"titleSize":14,"headerSize":12,"checkboxAndApplyColor":{"solid":{"color":"#
1A5276"}}}],"background":[{"color":{"solid":{"color":"#F5F7FA"}}}],"border":[{"color":{"solid":{"color":"#
2980B9"}},"show":true,"radius":8}],"visualHeader":[{"background":{"solid":{"color":"#D6EAF8"}},"foreground":{"solid":{"color":"#2C3E50"}},"border":{"solid":{"color":"#
2980B9"}}}],"visualTooltip":[{"themedTitleFontColor":{"solid":{"color":"#2C3E50"}},"themedValueFontColor":{"solid":{"color":"#2980B9"}},"actionFontColor":{"solid":{"color":"#
7F8C8D"}}}],"visualHeaderTooltip":[{"themedTitleFontColor":{"solid":{"color":"#
2C3E50"}}}],"filterCard":[{"$id":"Applied","backgroundColor":{"solid":{"color":"#D6EAF8"}},"foregroundColor":{"solid":{"color":"#
1A5276"}},"textSize":12,"transparency":0},{"$id":"Available","foregroundColor":{"solid":{"color":"#
1A5276"}},"textSize":12}]}},"page":{"*":{"background":[{"color":{"solid":{"color":"#FFFFFF"}},"transparency":100}],"outspace":[{"color":{"solid":{"color":"#EAF2F8"}}}]}}}}

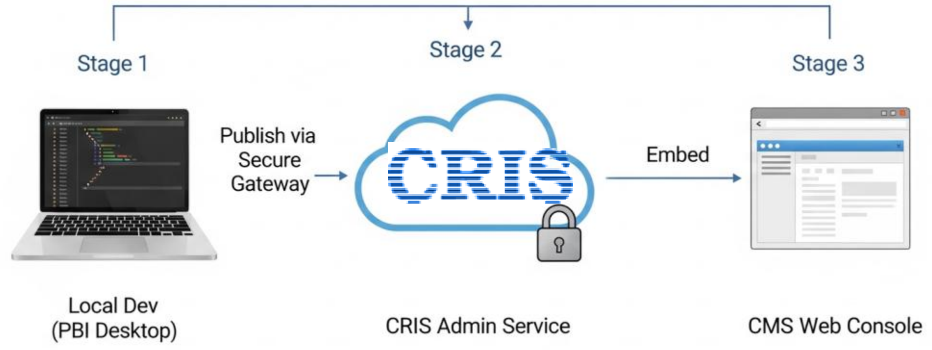
Deployment Architecture

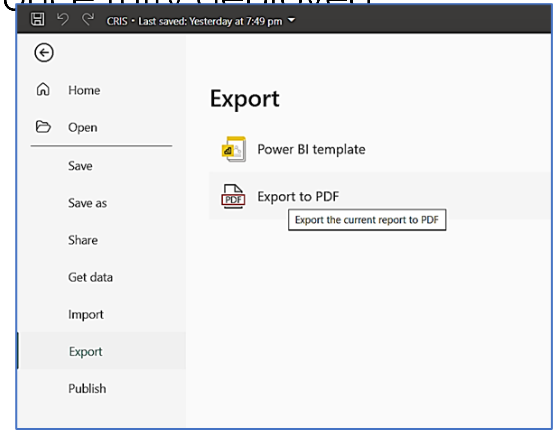
Hosting Environment: The CRIS Admin Power BI Service in a Microsoft Fabric Secure Env.

The Pipeline:

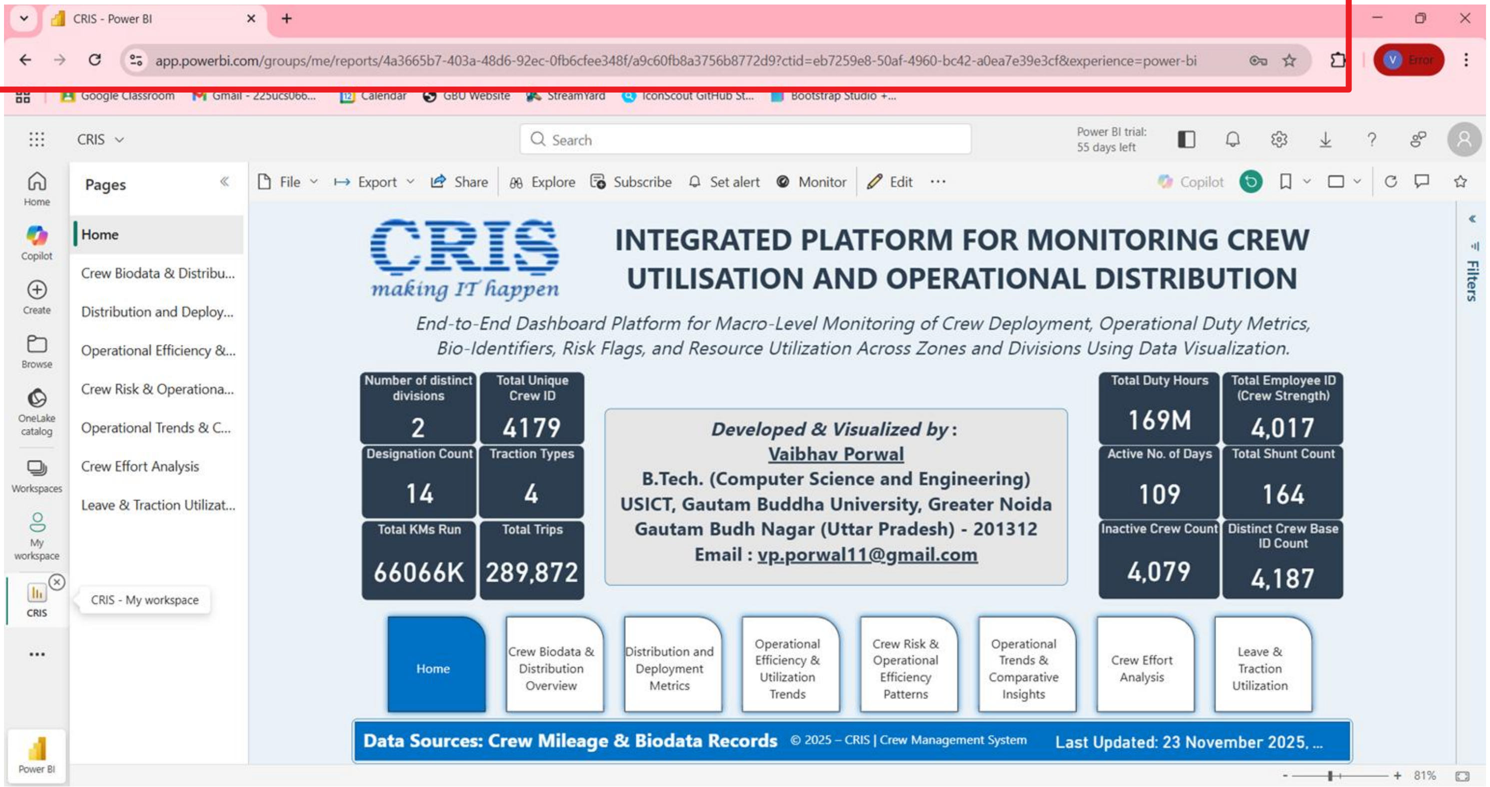
- Development: Local PBIX file (Current Stage).
- Publishing: Upload to CRIS Report Server.
- Access: Embedded URL integration into the CMS Web Portal.

Scalability: Designed to handle data from all 372 lobbies once fully deployed.





Dashboards on CRIS PowerBI Service



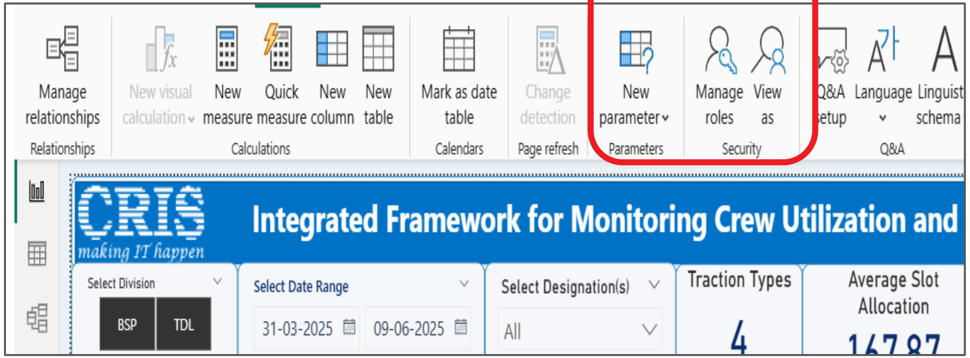
Data Security & Privacy

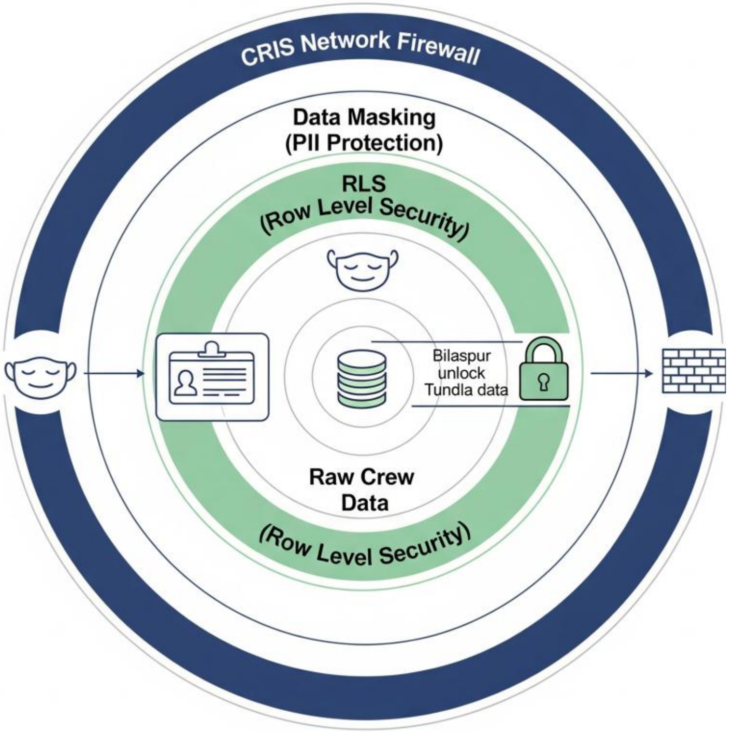
Audit Requirement: Addressing "IT Security" concerns raised in the CAG CRIS Report.

Data Masking: PII (Mobile Numbers, Employee IDs) is hidden in public views.

Row-Level Security (RLS):

- **Lobby User:** Sees only *Tundla* data.
- **Divisional User:** Sees *All Lobbies* in Division.
- **Zonal User:** Sees *All Divisions* in Zone.





Future Scope

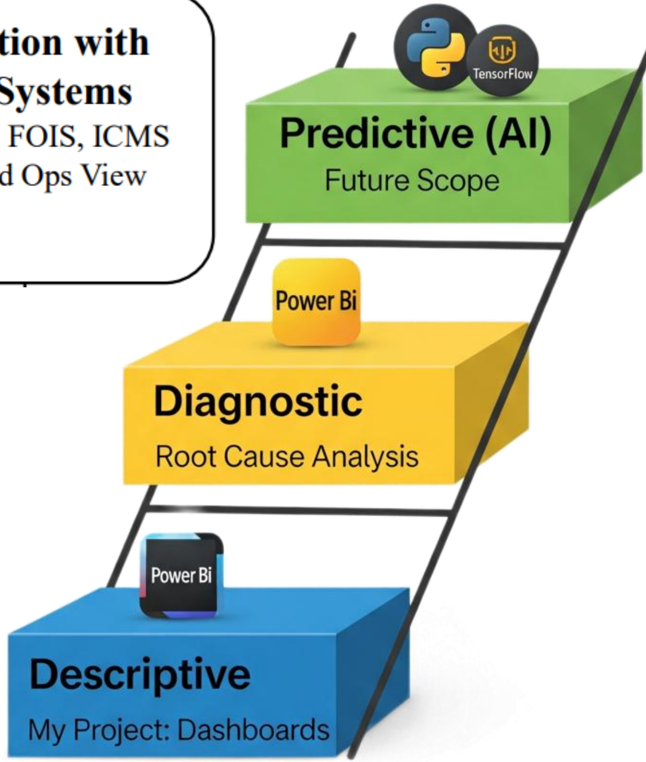
CURRENT IMPLEMENTATION
Data Integration & Macro-Level Dashboard
Crew Biodata + Mileage Analysis via PBIX

Direct CMS Database Connectivity
Automated Refresh from Backend Systems

Extension Across Indian Railways
Integration for Zonal & All Divisional HQ

Predictive & ML Capabilities
Crew Readiness, Fatigue & Anomaly Alerts

Integration with CRIS Systems
Crew COA, FOIS, ICMS for Unified Ops View



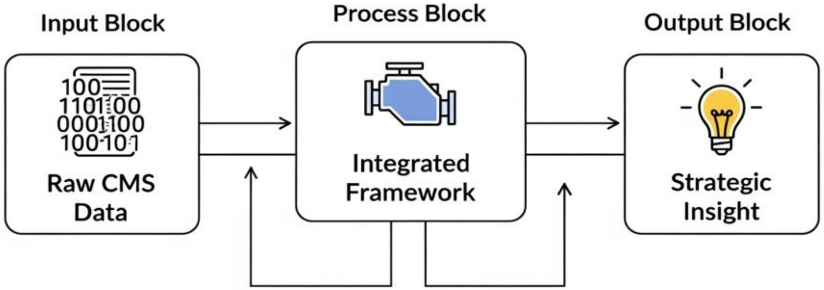
Conclusion

Successfully built an **end-to-end analytical laye**r integrated with extensive CMS ecosystem.

Key Impact Areas:

- **Operations**: Optimized deployment through spatial heatmap analysis.
- **Safety**: Enhanced visibility into Abnormality trends and Fatigue risks.
- **Finance**: Improved monitoring of Kilometer Allowance (KMA) and Overtime expenditures.
- **Automation**: Enabled consolidated crew insights and automated KPI generation

Final Verdict: Transformed raw CMS transaction logs into strategic dashboards.



Stakeholder Feedback & Remark

CRIS
making IT happen

रेलवे सूचना प्रणाली केन्द्र
(रेल मंत्रालय भारत सरकार का संगठन)

क्रिस
CRIS

CENTRE FOR RAILWAY INFORMATION SYSTEMS
(An Organisation of the Ministry of Railways, Govt. of India)

Date: 15.07.25

LETTER OF RECOMMENDATION

To Whom It May Concern,

It is with satisfaction that I write this letter in support of **Mr. Vaibhav Porwal**, who completed his Summer Practical Training at the **Centre for Railway Information Systems (CRIS)** under the **Crew Management System (CMS)** group, which I currently oversee as Chief Manager/CMS.

Vaibhav is a **third-year B.Tech. student** at the **University School of ICT, Gautam Buddha University, Greater Noida.** During his training, he worked on a project titled *"Integrated Framework for Monitoring Crew Utilization and Operational Distribution."* His responsibilities included analysing complex datasets related to crew operations, streamlining data structures, and delivering insights aligned with operational goals.

He demonstrated a clear understanding of the data and approached his tasks with maturity, clarity, and consistency. His contributions reflected sound technical thinking and a disciplined approach to data-driven analysis.

I recommend him for suitable academic or professional opportunities.

I also wish him continued success in his future pursuits.

For further information, I may be contacted at khanna.rachit@cris.org.in.

Sincerely,

15.7.25
RACHIT KHANNA
Chief Manager/RTIS
Centre For Railway Information Systems
Chanakyapuri, New Delhi-110021

Rachit Khanna
Chief Manager/CMS
Centre for Railway Information Systems (CRIS)
New Delhi

चाणक्यपुरी, नयी दिल्ली-110021
CHANAKYAPURI, NEW DELHI-110021
टेलीफोन/TELEPHONE : 24104525, 24106717 फैक्स/FAX : 91-11-26877893

"I write this in support of Vaibhav Porwal, who completed his Project Internship at CRIS under the CMS group, which I currently oversee as Chief Manager/CMS.

During his training, he worked on a project in which his responsibilities included analysing complex datasets related to crew operations, streamlining data structures, and delivering insights aligned with operational goals.

He demonstrated a clear understanding of the data and approached his tasks with maturity, clarity, and consistency. His contributions reflected sound technical thinking and a disciplined approach to data-driven analysis."

Rachit Khanna (IRSEE)
Chief Manager/ CMS
CRIS

Stakeholder Feedback & Remark

रेलवे सूचना प्रणाली केन्द्र
(रेल मंत्रालय भारत सरकार का संगठन)

CENTRE FOR RAILWAY INFORMATION SYSTEMS
(An Organisation of the Ministry of Railways, Govt. of India)

Date: 15.07.25

LETTER OF RECOMMENDATION

To Whom It May Concern,

I am pleased to write this letter in support of **Mr. Vaibhav Porwal**, who completed his **Summer Practical Training** under my guidance at the **Centre for Railway Information Systems (CRIS)** in the **Crew Management System (CMS)** group.

Vaibhav, currently a **3rd year B.Tech. student at the University School of ICT, Gautam Buddha University, Greater Noida**, undertook a project titled **"Integrated Framework for Monitoring Crew Utilization and Operational Distribution."** The project involved analysing and transforming operational crew data to extract patterns and insights for system-level decision-making.

His work required handling multi-source datasets, cleaning and structuring records, and developing analytical outputs relevant to performance monitoring and resource planning. Vaibhav approached the task with clarity, technical discipline, and a structured mindset. He worked independently, responded well to feedback, and maintained a professional and consistent work ethic throughout the training period.

I recommend him for future academic or professional opportunities. I also wish him the very best in all his endeavours.

For any further details, please feel free to contact me at **malik.praveen@cris.org.in**.

Sincerely,

Praveen Malik 15/07/2025

Praveen Malik
Principal Project Engineer (CMS)
Centre for Railway Information Systems (CRIS)
New Delhi

चाणक्यपुरी, नयी दिल्ली-110021
CHANAKYAPURI, NEW DELHI-110021
टेलीफोन/TELEPHONE : 24104525, 24106717 फैक्स/FAX : 91-11-26877893

"His work required handling multi-source datasets, cleaning and structuring records, and developing analytical outputs relevant to performance monitoring and resource planning. Vaibhav approached the task with clarity, technical discipline, and a structured mindset. He worked independently, responded well to feedback, and maintained a professional and consistent work ethic throughout the training period."

Praveen Malik
Principal Project Engineer/ CMS